

# SNA - college

Subject social network analysis - below i provide you a questions with answer but the work is vey complex

make it simple such that i can write in exam: Entity

Resolution:- Definition: Entity Resolution (ER) is the process of identifying, matching, and merging records that refer to the same real-world entity across one or more datasets, even when the records are slightly different. In social networks, this ensures that multiple accounts, profiles, or entries for the same person or organization are recognized as one entity. Why It's

Important: 1. Avoid duplicate data: Prevents counting

the same person multiple times. 2. Improve data

quality: Ensures accurate analytics and network

analysis. 3. Enhance recommendations: Correctly

identifies users to suggest connections, content, or

ads. 4. Enable proper community detection: Accurate

node identification improves clustering and community

analysis. Challenges in Entity Resolution: □ Different

naming conventions: e.g., "Manisha Das" vs. "M. Das". □

Incomplete data: Missing email, phone, or address. □

Typographical errors: Misspellings in names or

addresses. □ Multiple sources: Data coming from

different social media platforms or databases.

Techniques Used: 1. Rule-based matching: o Using

exact or partial matches on attributes (e.g., email,

phone number). 2. Probabilistic matching: o Assigns a

probability that two records represent the same entity.

3. Machine learning approaches: o Train models to

recognize patterns and predict if two records belong to

the same entity. 4. Graph-based methods: o Uses

network connections and relationships to resolve

entities. Example in Social Networks: □ Suppose a

person has two Facebook accounts: o Account 1:

Manisha Das, email: manisha@gmail.com o Account 2:

M. Das, email: manishadas@gmail.com □ Entity

resolution can detect that both accounts belong to the

same person and merge or link them for accurate

analysis.

Got it 👍 Let me simplify your answer for exam writing. Here's a short, clear version:

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## Entity Resolution (ER)

### Definition:

Entity Resolution is the process of finding and merging records that refer to the same person or organization, even if written differently. In social networks, it means recognizing multiple accounts of the same user as one entity.

### Importance:

1. Avoids duplicate data (no double counting).
2. Improves data quality (more accurate analysis).
3. Gives better recommendations (correctly links users).
4. Helps in proper community detection (accurate clusters).

### Challenges:

- Different naming styles: *"Manisha Das"* vs. *"M. Das"*.
- Incomplete data: missing email/phone.
- Typos: spelling mistakes.
- Multiple sources: data from many platforms.

### Techniques:

1. **Rule-based** → match by email, phone, etc.
2. **Probabilistic** → assign probability of being same.
3. **Machine learning** → train models to detect matches.
4. **Graph-based** → use network relations to resolve.

### Example:

Two accounts:

- *Manisha Das, email: manisha@gmail.com ↗*  
.....
  - *M. Das, email: manishadas@gmail.com ↗*  
.....
- ER identifies both as the same person and merges them.

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👍 This version is short, direct, and exam-friendly.

Do you want me to also **make bullet-point notes (super short, memory-friendly)** so you can recall quickly in exam hall?

## Link Prediction

### Definition:

Link Prediction is the task of predicting the likelihood of a future or missing connection between two nodes in a network based on existing network structure and node/edge attributes.

### Why Link Prediction is Important:

1. Friend suggestions: Recommending new friends to users in social networks.
2. Network completion: Detecting missing or hidden links in partially observed networks.
3. Understanding network evolution: Helps in studying how networks grow over time.

### Techniques for Link Prediction:

1. Similarity-based methods:
  - o Nodes with similar neighbors or attributes are more likely to connect.
  - o Example metrics:
    - Common neighbors: Nodes sharing many neighbors are likely to connect.
    - Jaccard coefficient: Measures the ratio of shared neighbors to total neighbors.
2. Probabilistic models:
  - o Estimate the probability of a link using statistical or machine learning models.

### Application Example:

- Facebook "People You May Know":
  - o Uses link prediction to suggest potential friends based on:
    - Mutual friends (common neighbors)
    - Shared interests or groups
    - Network activity

o This increases user engagement and helps expand the network.

Okay 👍 Let's simplify this into an exam-friendly version, like I did before:

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## Link Prediction

### Definition:

Link Prediction is the process of predicting whether two nodes in a network will form a connection in the future or if a missing link exists, using the current structure and attributes.

### Importance:

1. **Friend suggestions** (e.g., new friends on social media).
2. **Network completion** (find missing/hidden links).
3. **Understand network growth** (study how networks evolve).

### Techniques:

1. **Similarity-based** → nodes with similar neighbors/attributes connect more.
  - *Common neighbors*: more common friends → higher chance of link.
  - *Jaccard coefficient*: ratio of shared neighbors to total neighbors.
2. **Probabilistic models** → use statistics or ML to estimate link probability.

### Example (Facebook – “People You May Know”):

- Suggests friends using:
    - Mutual friends (common neighbors)
    - Shared interests/groups
    - User activity
  - Increases engagement and expands the network.
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👍 This version is short, easy to write, and straight to the point.

Want me to also prepare **super-short notes (1–2 line keywords only)** for *Link Prediction* just like “memory triggers” for quick recall in exam?

### 1. Example of a community in a social network.

Example of a community in a social network:

- A WhatsApp group of college friends – all members are connected, share messages, and interact frequently, forming a tightly-knit group within the larger WhatsApp network.

- Facebook group for photography enthusiasts – members share photos and tips.

- LinkedIn professional circle – a network of software engineers in AI sharing jobs and knowledge.

### 2. Community Discovery:

Definition:

Community discovery is the process of identifying groups of nodes in a network that are more densely connected to each other than to the rest of the network. These groups are called communities or clusters.

- Communities often share common interests, roles, or attributes.

- Detecting communities helps understand the organization and structure of networks.

- Algorithms for community detection include:

- o Modularity-based methods (maximize intra-community edges)
- o Hierarchical clustering (build tree-like structure)
- o Label propagation (nodes adopt majority labels of neighbors)

Example:

- In a Facebook network, a community could be a group of college friends who interact frequently.

### 3. Communities in Context:

Definition:

Communities in context refers to analyzing communities with respect to external information or real-world context. It connects the detected communities to real-life roles, topics, or functions.

Why it matters:

□ Helps interpret the meaning behind communities.  
□ Useful in marketing, recommendation systems, and understanding social behavior.  
□ Communities are not just clusters of nodes—they often represent real-world structures or shared interests.  
Example:  
□ On LinkedIn, a detected community could be a group of software engineers working in AI. Contextual information (job role, skills) explains why this community exists.  
Benefits of Community Discovery in SNA:  
1. Identify influential groups for targeted marketing.  
2. Understand social behavior and information flow.  
3. Detect hidden patterns or subgroups within large networks.  
4. Improve recommendation systems by suggesting relevant content or connections.

Perfect 👍 Let's make this exam-friendly, short, and easy to remember.

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## 1. Example of a Community in a Social Network

- **WhatsApp group** of college friends (close interactions).
  - **Facebook group** for photography (share photos, tips).
  - **LinkedIn circle** of AI engineers (share jobs, knowledge).
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## 2. Community Discovery

**Definition:** Finding groups of nodes that are more connected with each other than with the rest of the network.

- Communities = groups with common interests/roles.
- Helps understand structure of networks.

**Algorithms:**

1. **Modularity-based** → maximize internal connections.

2. **Hierarchical clustering** → tree-like structure.
3. **Label propagation** → nodes adopt neighbors' labels.

**Example:**

Facebook college friends group detected as a community.

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### 3. Communities in Context

**Definition:** Connecting detected communities to real-world meaning (interests, roles, topics).

**Why it matters:**

- Explains *why* communities exist.
- Useful in marketing, recommendations, behavior analysis.

**Example:**

On LinkedIn → AI engineers form a community due to job roles/skills.

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### Benefits of Community Discovery

1. Identify influential groups → marketing.
  2. Understand behavior & information flow.
  3. Detect hidden patterns/subgroups.
  4. Improve recommendations.
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👉 This way, you can **write full answers in short time** and also **revise quickly** before exams.

Do you want me to also make a **super-condensed 1-page notes version (only key terms + keywords)** for these three topics so you can revise at a glance?